```
> restart;#Flux Pto force aprox;
> de := 1/V^2·H(r)·diff(u(t), t, t) + u(t)·diff(H(r), r, r) = 0;
```

$$de := \frac{H(r)\left(\frac{d^2}{dt^2}\,u(t)\right)}{V^2} + u(t)\left(\frac{d^2}{dr^2}\,H(r)\right) = 0 \tag{1}$$

```
> dsolve(de, u(t));
```

$$u(t) = _C1 \sin\left(\frac{V\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) + _C2 \cos\left(\frac{V\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) \tag{2}$$

```
> eq1 := r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r) = 0;
```

$$eq1 := r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r) = 0 \tag{3}$$

```
> eq2 := (r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r))/r = 0;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r)}{r} = 0 \tag{4}$$

```
> # Thus with R:=3.566·10^22 m  we can  ignore the therm -b^2·r/R for
  1/R < r < R;
```

$$\frac{1}{R} < r < R \tag{5}$$

```
> 1/r diff((r·diff H(r), r), r) + ν^2· H(r) = 0;
```

$$\frac{2\,diff\left(\frac{d}{dr}\,H(r)\right) + r\,diff\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + \nu^2 H(r) = 0 \tag{6}$$

```
> de1 := 1/r·diff(r·diff(H(r), r), r) + ν^2·H(r);
```

$$de1 := \frac{\frac{d}{dr}\,H(r) + r\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + \nu^2 H(r) \tag{7}$$

```
> bc := H(2 R) = 0, D(H)(1/R) = R ;
```

$$bc := H(2\,R) = 0, \mathrm{D}(H)\left(\frac{1}{R}\right) = R \tag{8}$$

```
> dsol := dsolve({de1, bc}, H(r));
```

$$dsol := H(r) = \frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}(0, \nu\,r)}{\nu\left(\mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\,\mathrm{BesselY}(0, \nu\,r)}{\nu\left(\mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right)\right)} \tag{9}$$

```
> H(r) := -(R BesselY(0, 2 ν R) BesselJ(0, ν r)) / (ν (BesselJ(1, ν/R) BesselY(0, 2 ν R) - BesselY(1, ν/R) BesselJ(0, 2 ν R)));+
```

$$H := r \mapsto -\frac{R\cdot\mathrm{BesselY}(0, 2\cdot\nu\cdot R)\cdot\mathrm{BesselJ}(0, \nu\cdot r)}{\nu\cdot\left(\mathrm{BesselY}(0, 2\cdot\nu\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\cdot\nu\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)} + \frac{\mathrm{BesselJ}(0, 2\cdot\nu\cdot R)\cdot R\cdot\mathrm{BesselY}(0, \nu\cdot r)}{\nu\cdot\left(\mathrm{BesselY}(0, 2\cdot\nu\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\cdot\nu\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)} \tag{10}$$

```
> u(t) := _C1 sin(V sqrt(d^2/dr^2 H(r)) t / sqrt(H(r))) + _C2 cos(V sqrt(d^2/dr^2 H(r)) t / sqrt(H(r)));
```

$$u := t \mapsto _C1\cdot\sin\left(\frac{V\cdot\sqrt{\frac{d^2}{dr^2}H(r)}\cdot t}{\sqrt{H(r)}}\right) + _C2\cdot\cos\left(\frac{V\cdot\sqrt{\frac{d^2}{dr^2}H(r)}\cdot t}{\sqrt{H(r)}}\right) \tag{11}$$

```
> Hpto := subs(H(r) = -(R BesselY(0, 2 ν R) BesselJ(0, ν r)) / (ν (BesselJ(1, ν/R) BesselY(0, 2 ν R) - BesselY(1, ν/R) BesselJ(0, 2 ν R))) + (BesselJ(0, 2 ν R) R BesselY(0, ν r)) / (ν (BesselJ(1, ν/R) BesselY(0, 2 ν R) - BesselY(1, ν/R) BesselJ(0, 2 ν R))), H(r) = u(t)/r) ;# Because we have Effective Wave Function
```

$$Hpto := \frac{1}{r}\left(\left(-\frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)} \right.\right. \tag{12}$$

$$\left.\left. + \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\,\mathrm{BesselY}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)} \right)$$

$$\left(_C1\,\sin\left(\left(V \right.\right.\right.$$

$$\left(\frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\left(\mathrm{BesselJ}(0, \nu\,r) - \frac{\mathrm{BesselJ}(1, \nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)} \right.$$

$$\left.\left.\left. - \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\left(\mathrm{BesselY}(0, \nu\,r) - \frac{\mathrm{BesselY}(1, \nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)} \right)^{1/2} t \right) \Big/$$

$$\left(-\frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)}\right.$$

$$\left. + \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\,\mathrm{BesselY}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)}\right)^{1/2}\Bigg)$$

$$+ _C2\cos\Bigg(\Bigg(V$$

$$\left(\frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\left(\mathrm{BesselJ}(0, \nu\,r) - \frac{\mathrm{BesselJ}(1, \nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)}\right.$$

$$\left. - \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\left(\mathrm{BesselY}(0, \nu\,r) - \frac{\mathrm{BesselY}(1, \nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)}\right)^{1/2} t\Bigg)\Bigg/$$

$$\left(-\frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)}\right.$$

$$\left. + \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\,R\,\mathrm{BesselY}(0, \nu\,r)}{\nu\left(\mathrm{BesselY}(0, 2\,\nu\,R)\,\mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\,\mathrm{BesselY}\left(1, \frac{\nu}{R}\right)\right)}\right)^{1/2}\Bigg)\Bigg)\Bigg)$$

```
> FPto0 := subs( t = 0, Hpto / (2·Pi·r·l) );
```

(13)

$$FPto0 := \frac{1}{2\,r^2\,\pi\,l}\left(\left(\right.\right.$$

$$-\frac{R\,\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}$$

$$+\frac{\mathrm{BesselJ}(0,2\,\nu\,R)\,R\,\mathrm{BesselY}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}\left.\right)$$

$$(_C1\,\sin(0)+_C2\,\cos(0))\left.\right)$$

(13)

```
> solve(FPto=0);
```

$$0$$

(14)

```
> FPto := subs(t=1, _C1=1, _C2=1, k=1, Hpto/(2·Pi·r·l));
```

$$FPto := \frac{1}{2\,r^2\,\pi\,l}\left(\left(\right.\right.$$

$$-\frac{R\,\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}$$

$$+\frac{\mathrm{BesselJ}(0,2\,\nu\,R)\,R\,\mathrm{BesselY}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}\left.\right)$$

(15)

$$\left(\sin\left(\left(\left(V \left(\left(R\,\mathrm{BesselY}(0, 2\,\nu\,R) \left(\mathrm{BesselJ}(0, \nu\,r) - \frac{\mathrm{BesselJ}(1, \nu\,r)}{\nu\,r} \right) \nu \right) \Big/ \right. \right. \right. \right. \right.$$

$$\left(\mathrm{BesselY}(0, 2\,\nu\,R)\ \mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\ \mathrm{BesselY}\left(1, \frac{\nu}{R}\right) \right)$$

$$\left. \left. \left. \left. \left. - \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\ R \left(\mathrm{BesselY}(0, \nu\,r) - \frac{\mathrm{BesselY}(1, \nu\,r)}{\nu\,r} \right) \nu}{\mathrm{BesselY}(0, 2\,\nu\,R)\ \mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\ \mathrm{BesselY}\left(1, \frac{\nu}{R}\right)} \right)^{1/2} \right) \Big/ \right.$$

$$\left(- \frac{R\,\mathrm{BesselY}(0, 2\,\nu\,R)\ \mathrm{BesselJ}(0, \nu\,r)}{\nu \left(\mathrm{BesselY}(0, 2\,\nu\,R)\ \mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\ \mathrm{BesselY}\left(1, \frac{\nu}{R}\right) \right)} \right.$$

$$\left. \left. + \frac{\mathrm{BesselJ}(0, 2\,\nu\,R)\ R\,\mathrm{BesselY}(0, \nu\,r)}{\nu \left(\mathrm{BesselY}(0, 2\,\nu\,R)\ \mathrm{BesselJ}\left(1, \frac{\nu}{R}\right) - \mathrm{BesselJ}(0, 2\,\nu\,R)\ \mathrm{BesselY}\left(1, \frac{\nu}{R}\right) \right)} \right)^{1/2} \right)$$

$$+\cos\left(\left(V\right.\right.$$

$$\left(\frac{R\,\mathrm{BesselY}(0,2\,\nu\,R)\left(\mathrm{BesselJ}(0,\nu\,r)-\frac{\mathrm{BesselJ}(1,\nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)}\right.$$

$$\left.\left.-\frac{\mathrm{BesselJ}(0,2\,\nu\,R)\,R\left(\mathrm{BesselY}(0,\nu\,r)-\frac{\mathrm{BesselY}(1,\nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)}\right)^{1/2}\right)\Big/$$

$$\left(-\frac{R\,\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}\right.$$

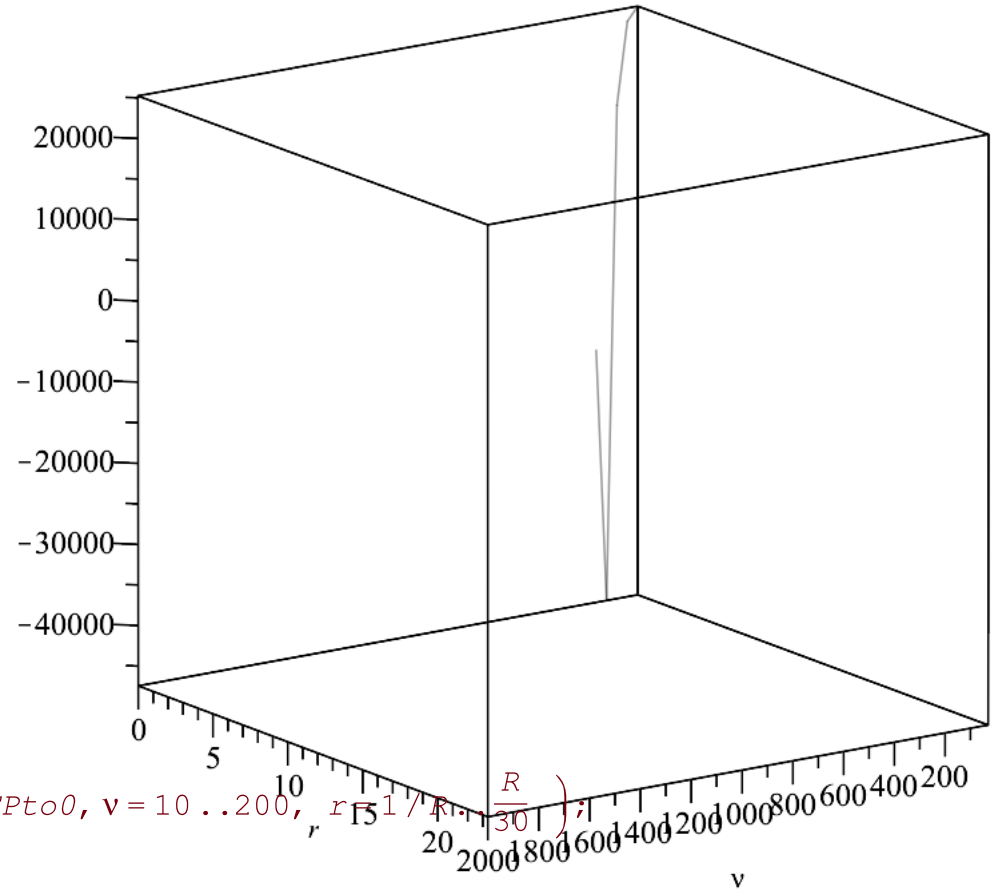
$$\left.\left.+\frac{\mathrm{BesselJ}(0,2\,\nu\,R)\,R\,\mathrm{BesselY}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}(0,2\,\nu\,R)\,\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)-\mathrm{BesselJ}(0,2\,\nu\,R)\,\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\right)}\right)^{1/2}\right)\right)\right)$$

```
> R := 7·10^2;  _C1 := 1; _C2 := 1; V := 3000;  l := 8;
```

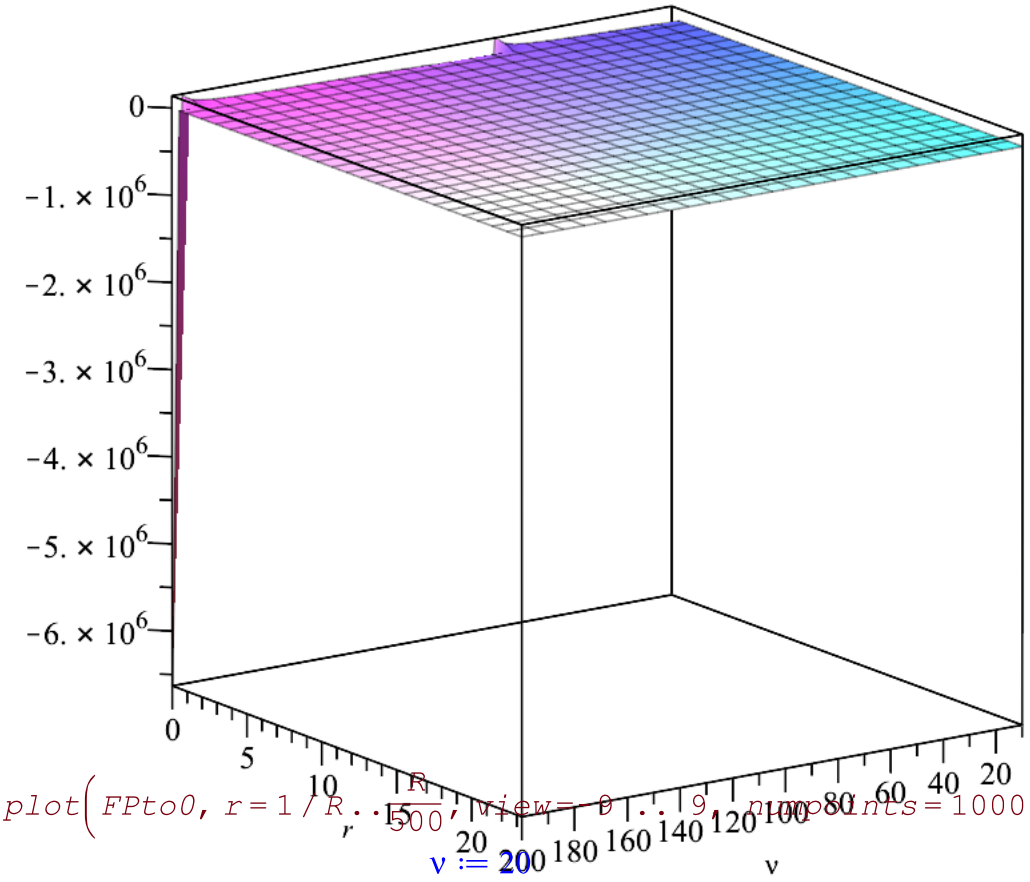
$$R := 700$$
$$_C1 := 1$$
$$_C2 := 1$$
$$V := 3000$$
$$l := 8$$

(16)

```
> plot3d( FPto, ν = 30..2000,  r = 1/R..R/30 );
```

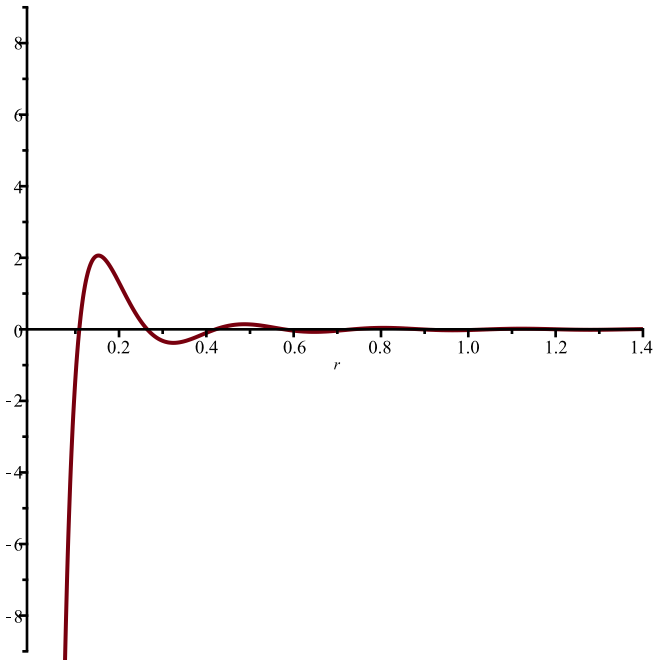



```
> plot3d( FPto0, ν = 10 ..200, r = 1/R..R/30 );
```



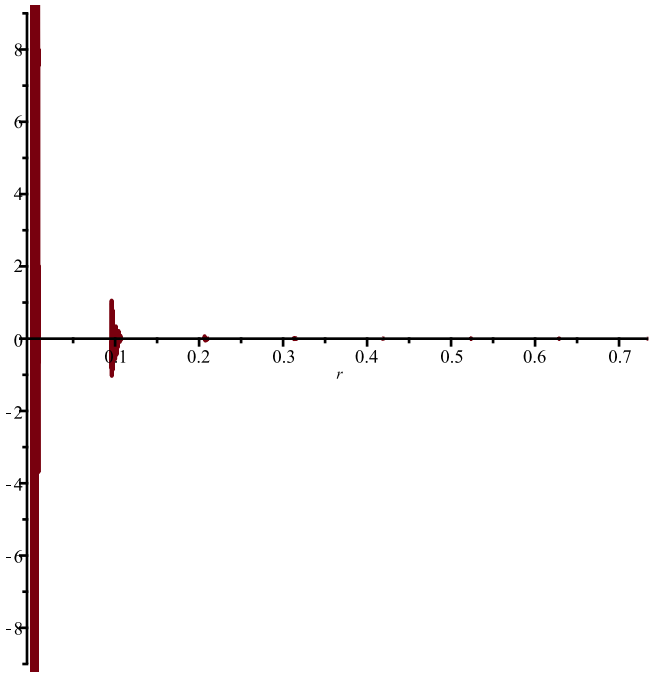

```
> ν := 20; plot( FPto0, r = 1 / R .. 1/500, view= 0 .. 9, numpoints = 1000 );
```

ν := 20




```
> ν := 30; plot( FPto, r = 1 / R .. R/900, view=-9 .. 9, numpoints = 1000 );
```

$\nu := 30$



>